

Stefano Ghislandi

Curriculum Vitæ

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Personal Data

Birth Seriate (Italy), February 25th, 1996
Citizenship Italian

Scientific Interests

- Neutrinoless double beta decay searches
- Neutrino physics and astrophysics
- Low radioactivity techniques
- Cryogenic detectors development and cryogenic systems
- Monte Carlo simulations

Education

Apr 2025 – Present **Postdoctoral Associate**, at *Massachusetts Institute of Technology (MIT)*

2020–2025 **Astroparticle Ph.D. School**, at *Gran Sasso Science Institute (GSSI)*

2018–2020 **Master Degree in Particle Physics**, *Università degli Studi di Milano-Bicocca*, (Italy), 110/110 magna cum laude, weighted exam average: 29.9/30
Thesis: On the Background Reduction with Organic Compounds and LAr in Rare Events Experiments with Thermal Detectors

2015–2018 **Bachelor Degree in Physics**, *Università degli Studi di Milano-Bicocca*, (Italy), 110/110 magna cum laude, weighted exam average: 29/30
Thesis: A Bi-dimensional Implementation in the (E, ω) Space of the Optimal Filter for Thermal Detectors

2010–2015 **High School Degree**, *Liceo Scientifico F.Lussana*, Bergamo, 97/100

Outreach

2021–2024 **Scientific Committee Member**, of the *Premio Asimov*

Mar–Jun 2024 **Member of the Local Organizing Committee**, of the *IDM 2024 conference*

30 Sep 2022 **Sharper 2022**, *researcher night*, L'Aquila

2022 – Present **Contributions to INFN-LNGS CUPID public webpage**

24 Sep 2021 **Sharper 2021**, *researcher night*, L'Aquila

A.Y. **Physics I teaching tutor**, (60 hrs job as type C tutor) at Università degli Studi di Milano-Bicocca
2019–2020

Schools and Workshops

- **International School of Subnuclear Physics (ISSP EMFCSC) 2023**, Erice
- **CryoCourse 2022**, *European intensive course on advanced Cryophysics and Cryogenics*, Heidelberg
- **INFN School of Statistics 2022**, Paestum
- **FNHP 2021**, *Frontiers in Nuclear and Hadronic Physics 2021*, Galileo Galilei Institute (GGI)
- **Code/Astro 2021: A Software Engineering Workshop for Astronomy**
- **ISAPP 2021 Valencia: Neutrino Physics, Astrophysics and Cosmology**
- **CERN Summer Student Lectures 2019**, *consisting of 1 month lectures about particle physics, during my internship at CERN*

Talk and Posters

- **Contributed Talk at LTD 2025** about the latest results from the CUORE experiment.
- **Invited seminar at Yale** about precision spectral shape measurements of β decays with cryogenic calorimeters.
- **Poster at Neutrino 2024** about the results of the CUORE background model, part of my thesis work.
- **Parallel Talk at TAUP 2023** about the first results on the CUORE background model, part of my thesis work, Vienna.
- **Talk at Rencontres du Vietnam, Windows on the Universe 2023** about the latest CUORE and CUPID results, Quy Nhon - Vietnam.
- **Talk at ISSP EMFCSC 2023** about the CUORE background model, Erice - Italy
- **Talk at IFAE 2023** about the Res Nova proposed experiment (*best talk in Section "New Technologies"*), Catania.
- **Talk at PSI2022** about the latest CUORE results, Baden - Switzerland.
- **Poster at CryoCourse 2022** about the ACCESS project, Heidelberg.
- **Poster at Neutrino 2022** about my work on the CUORE background model, Virtual.
- **Talk at XIX International Workshop on Neutrino Telescopes** about my Master Thesis work, Virtual. DOI 10.5281/zenodo.5744979.
- **Talk at SIF 106th National Congress** about my Master Thesis work, Virtual.

Awards, Scholarships

- **Best poster** at the conference Neutrino 2024.

- **Giovanni Polvani prize** of Euro 1.000 awarded from the Italian Physics Society (SIF) for scientific industriousness.
- **Oreste Piccioni Diploma** at the International School of Subnuclear Physics (ISSP EMFCSC) 2023, Erice - Italy.
- **Best talk** in section "New Technologies" at the conference IFAE 2023.
- **Selected for Ph.D. position** at the Gran Sasso Science Institute, 2020.
- **Talk at SIF 2020 awarded with publication** on *Il Nuovo Cimento C*.
- **Selected for CERN Summer Student Scholarship**, 2019 .

Review Activities

- Nuclear Instruments and Methods in Physics Research Section A

Publications, Preprints and Proceeding

36. CUORE collaboration
Exploring the keV-scale physics potential of CUORE
Arxiv; doi: 10.48550/arXiv.2505.23955
35. CUORE collaboration
Reconstruction of cosmic-ray muon events with CUORE
Arxiv; doi: 10.48550/arXiv.2509.05528
34. CUORE collaboration
End-to-End Data Analysis Methods for the CUORE Experiment
Arxiv; doi: 10.48550/arXiv.2510.25720
33. RES-NOVA collaboration
Probing dark matter interactions with a RES-NOVA prototype cryogenic detector
Arxiv; doi: 10.48550/arXiv.2601.16251
32. CUPID collaboration
CUPID, the Cuore upgrade with particle identification
Eur. Phys. J. C 85, 737 (2025); doi: 10.1140/epjc/s10052-025-14352-1
31. RES-NOVA collaboration
New dark matter direct search based on archaeological Pb
Phys. Rev. D 111, 103050; doi: 10.1103/wc vd-rk1f
30. CUORE Collaboration
The detection of marine microseismic activity with the CUORE tonne-scale cryogenic experiment
Comm. Physics; doi: arXiv.2505.09652
29. CUPID Collaboration
Sensitivity of the CUPID experiment to $0\nu\beta\beta$ decay of ^{100}Mo
Accepted by Eur. Phys. J. C; doi: arXiv.2504.14369
28. CUPID Collaboration
A gravity-based mounting approach for large-scale cryogenic calorimeter arrays.
Eur. Phys. J. C 85, 935 (2025); doi: 10.1140/epjc/s10052-025-14613-z

28. CUORE Collaboration
Half-life and precision shape measurement of $2\nu\beta\beta$ decay of ^{130}Te
Phys. Rev. Lett. 135, 082501; doi: 10.1103/jdhf-hn4l
27. RES-NOVA Collaboration
New dark matter direct search based on archaeological Pb
Phys. Rev. D 111, 103050 (2025); doi: 10.1103/wcvd-rk1f
26. CUORE Collaboration
Search for fractionally charged particles with CUORE
Phys. Rev. Lett. 133, 241801 (2024); 10.1103/PhysRevLett.133.241801
25. CUORE Collaboration
Data-driven background model for the CUORE experiment
Phys. Rev. D 110, 052003 (2024); doi: 10.1103/PhysRevD.110.052003
24. CUORE Collaboration
Constraints on Lepton Number Violation with the 2 tonne-yr CUORE Dataset
Science 390 (2025) 6777, 1029-1032; 10.1126/science.adp6474
23. L. Pagnanini, G. Benato, P. Carniti, E. Celi, D. Chiesa, J. Corbett, I. Dafinei, S. DiDomizio, P. DiStefano, **S. Ghislandi**, C. Gotti, D. L. Helis, R. Knobel, J. Kosten-salo, J. Kotila, S. Nagorny, G. Pessina, S. Pirro, S. Pozzi, A. Puiu, S. Quitadamo, M. Sisti, J. Suhonen and S. Kuznetsov
Simultaneous Measurement of Half-Life and Spectral Shape of ^{115}In β -decay with an Indium Iodide Cryogenic Calorimeter
Phys. Rev. Lett. 133, 122501 (2024); doi: 10.1103/PhysRevLett.133.122501
22. **S. Ghislandi** and S. Pozzi on behalf of the CUORE collaboration
First results from the CUORE background model
PoS TAUP2023 (2024) 174; doi: 10.22323/1.441.0174
21. CUPID and CROSS Collaborations
Twelve-crystal prototype of Li_2MoO_4 scintillating bolometers for CUPID and CROSS experiments
2023 JINST 18 P06018; doi: 10.1088/1748-0221/18/06/P06018
20. S. Dell'Oro, A. Formicola, **S. Ghislandi**, L. Ioannucci, M. Laubenstein, B. Lehnert, S. Nagorny, S. Nisi, L. Pagnanini
Deep-underground search for the decay of $^{180\text{m}}\text{Ta}$ with an ultra-low-background HPGe detector
Eur. Phys. J. C 83, 925 (2023); doi: 10.1140/epjc/s10052-023-12087-5
19. CUPID Collaboration
A first test of CUPID prototypal light detectors with NTD-Ge sensors in a pulse-tube cryostat
2023 JINST 18 P06033; doi: 10.1088/1748-0221/18/06/P06033
18. L. Pagnanini, G. Benato, P. Carniti, E. Celi, D. Chiesa, J. Corbett, I. Dafinei, S. DiDomizio, P. DiStefano, **S. Ghislandi**, C. Gotti, D. L. Helis, R. Knobel, J. Kosten-salo, J. Kotila, S. Nagorny, G. Pessina, S. Pirro, S. Pozzi, A. Puiu, S. Quitadamo, M. Sisti, and J. Suhonen
Array of Cryogenic Calorimeters to Evaluate the Spectral Shape of forbidden β -decays: the ACCESS project
Eur. Phys. J. Plus 138, 445 (2023); doi: 10.1140/epjp/s13360-023-03946-x

17. Biassoni, M., Brofferio, C., Faverzani, M., Ferri, E., Nutini, I., Pettinacci, V., Pozzi, S., **Ghislandi, S.**, Quitadamo, S.
Towards the development of a polymer-based assembly for cryogenic detectors for neutrinoless double beta decay
Eur. Phys. J. Plus 138, 384 (2023); 10.1140/epjp/s13360-023-03991-6
16. CUPID Collaboration
CUPID: The Next-Generation Neutrinoless Double Beta Decay Experiment
J Low Temp Phys (2022); doi: 10.1007/s10909-022-02909-3
15. CUORE Collaboration
Latest Results from the CUORE Experiment
J Low Temp Phys 209,927–935 (2022); doi: 10.1007/s10909-022-02873-y
14. CUORE Collaboration
New direct limit on neutrinoless double beta decay half-life of ^{128}Te with CUORE
Phys. Rev. Lett. 129, 222501; doi: Phys. Rev. Lett. 129, 222501
13. G. Benato, M. Biassoni, C. Brofferio, E. Celi, S. Dell'Oro, A. Drobyzhev, A. Gianvecchio, M. Girola, **S. Ghislandi**, Y. G. Kolomensky, I. Nutini, M. Olmi, L. Pagnanini, A. Puiu, S. Quitadamo
Development of ultralow-background cryogenic calorimeters for the measurement of surface α contamination
Appl. Radiat. Isotopes 193, 110681; 10.1016/j.apradiso.2023.110681
12. RES-NOVA group of interest
Characterization of a large mass archaeological lead-based cryogenic detectors for the RES-NOVA experiment
Appl. Radiat. Isotopes, 194, 110704; 10.1016/j.apradiso.2023.110704;
11. CUORE Collaboration
An Energy-dependent Electro-thermal Response Model of CUORE Cryogenic Calorimeter
JINST 17 P11023 (2022); doi: 10.1088/1748-0221/17/11/P11023
10. CUORE Collaboration
Search for Neutrinoless β^+EC Decay of ^{120}Te with CUORE
Phys. Rev. C 105, 065504 (2022); doi: 10.1103/PhysRevC.105.065504
9. M. Biassoni, C. Brofferio, P. Carniti, M. Faverzani, E. Ferri, **S. Ghislandi**, A. Giachero, C. Gotti, I. Nutini, G. Pessina, S. Quitadamo
Pulse Shape Analysis in Low-temperature Calorimeters Read-Out by NTD Thermistors
J. Low. Temp. Phys. (2022); doi: 10.1007/s10909-022-02822-9;
8. RES-NOVA group of interest
Radiopurity of a kg-scale PbWO_4 cryogenic detector produced from archaeological Pb for the RES-NOVA experiment
Eur. Phys. J. C 82, 692 (2022); doi : 10.1140/epjc/s10052-022-10656-8;
7. CUORE Collaboration
Expected sensitivity to ^{128}Te neutrinoless double beta decay with the CUORE TeO_2 cryogenic bolometers
J. Low. Temp. Phys. (2022); doi: 10.1007/s10909-022-02738-4

6. CUPID Collaboration
Optimization of the first CUPID detector module
Eur. Phys. J. C, 82 9 (2022) 810 doi: 10.1140/epjc/s10052-022-10720-3
5. CUORE collaboration
CUORE opens the door to tonne-scale cryogenics experiments
Prog. Part. Nucl. Phys. 122, 103902 (2022); doi: 10.1016/j.pnpnp.2021.103902
4. M. Biassoni, C. Brofferio, P. Carniti, M. Faverzani, E. Ferri, **S. Ghislandi**, A. Giachero, C. Gotti, I. Nutini, G. Pessina, S. Quitadamo
A Thermal Model for Low Temperature TeO₂ Calorimeters
J. Low. Temp. Phys. 206, 80–96 (2022); doi: 10.1007/s10909-021-02639-y;
3. M. Biassoni, C. Brofferio, M. Faverzani, E. Ferri, **S. Ghislandi**, S. Milana, I. Nutini, V. Pettinacci, S. Pozzi and S. Quitadamo
An acrylic assembly for low-temperature detectors
Eur. Phys. J. Plus 136, 986 (2021); doi: 10.1140/epjp/s13360-021-01978-9
2. CUPID Collaboration
Searching for New Physics in two-neutrino double beta decay with CUPID
J.Phys.Conf.Ser. 2156 (2021); doi: 10.1088/1742-6596/2156/1/012233
1. **S. Ghislandi**, M. Biassoni, I. Nutini, S. Pozzi and S. Quitadamo
Organic compounds for rare events physics
Nuovo Cim.C 44, 2-3, 93 (2021) doi: 10.1393/ncc/i2021-21093-6

Computing Skills

Operative Systems	Unix/Linux, Windows (basic knowledge)
Programming Languages	Working knowledge: C++, Python, Bash, SQL Basic knowledge: C, MATLAB, CMake
Typesetting	L ^A T _E X, Markdown
Other Software	ROOT, RooFit, RooUnfold, JAGS (basics), BAT, Geant4, GitHub

Languages

Italian	Mothertongue
English	Fluent
French	Basic